



COIMBATORE INSTITUTE OF TECHNOLOGY

GUARDAI – NEXT GENERATION FINANCIAL FRAUD DETECTION

MINI PROJECT REVIEW

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PROBLEM STATEMENT

WHY TRADITIONAL FRAUD DETECTION IS FAILING?

1. SOPHISTICATED FRAUD RINGS

- FAKE ACCOUNTS
- SHARED DEVICES
- SHARED IPS
- SYNTHETIC IDENTITIES



2. IMBALANCE PROBLEM

- FRAUD TRANSACTIONS ARE VERY FEW
- LESS THAN 0.5% OF TOTAL DATA
- NORMAL ML MODELS FAIL



■ LEGITIMATE (> 99.5%) ■ FRAUD (< 0.5%)

NORMAL ML MODELS GET BIASED TOWARDS MAJORITY CLASS AND FAIL TO DETECT FRAUD.

3. BLACK BOX PROBLEM

- TRADITIONAL AI CANNOT EXPLAIN DECISIONS
- BANKS NEED EXPLANATIONS



BANKS REQUIRE CLEAR EXPLANATIONS FOR REGULATORY COMPLIANCE.



LACK OF EXPLAINABILITY REDUCES TRUST IN AI SYSTEMS.

LITERATURE SURVEY

SOLUTION OVERVIEW

DATA SOURCES



USERS



DEVICES

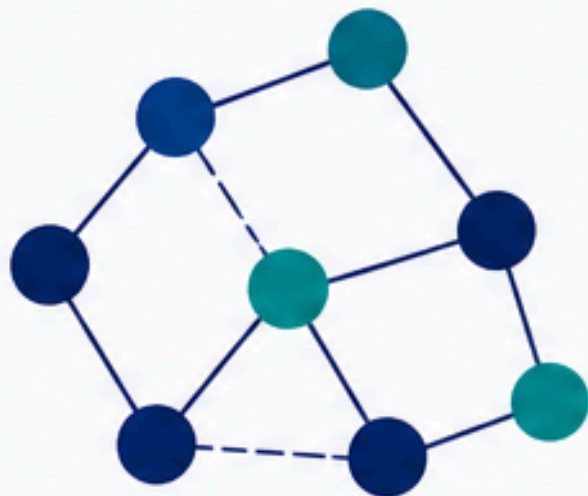


IP ADDRESSES



TRANSACTIONS

GRAPH CONSTRUCTION



NODES: Users /
Devices / Transactions

EDGES: Shared IPs /
Devices / Interactions

HYBRID AI ENGINE

GNN MODELS

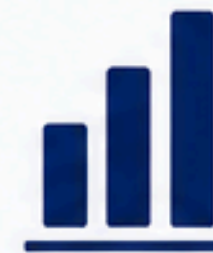


+

XGBOOST



EXPLAINABLE OUTPUT



SHAP
EXPLANATIONS



GNNEXPLAINER
VISUALIZATIONS

ACTIONS



FRAUD
PREDICTION

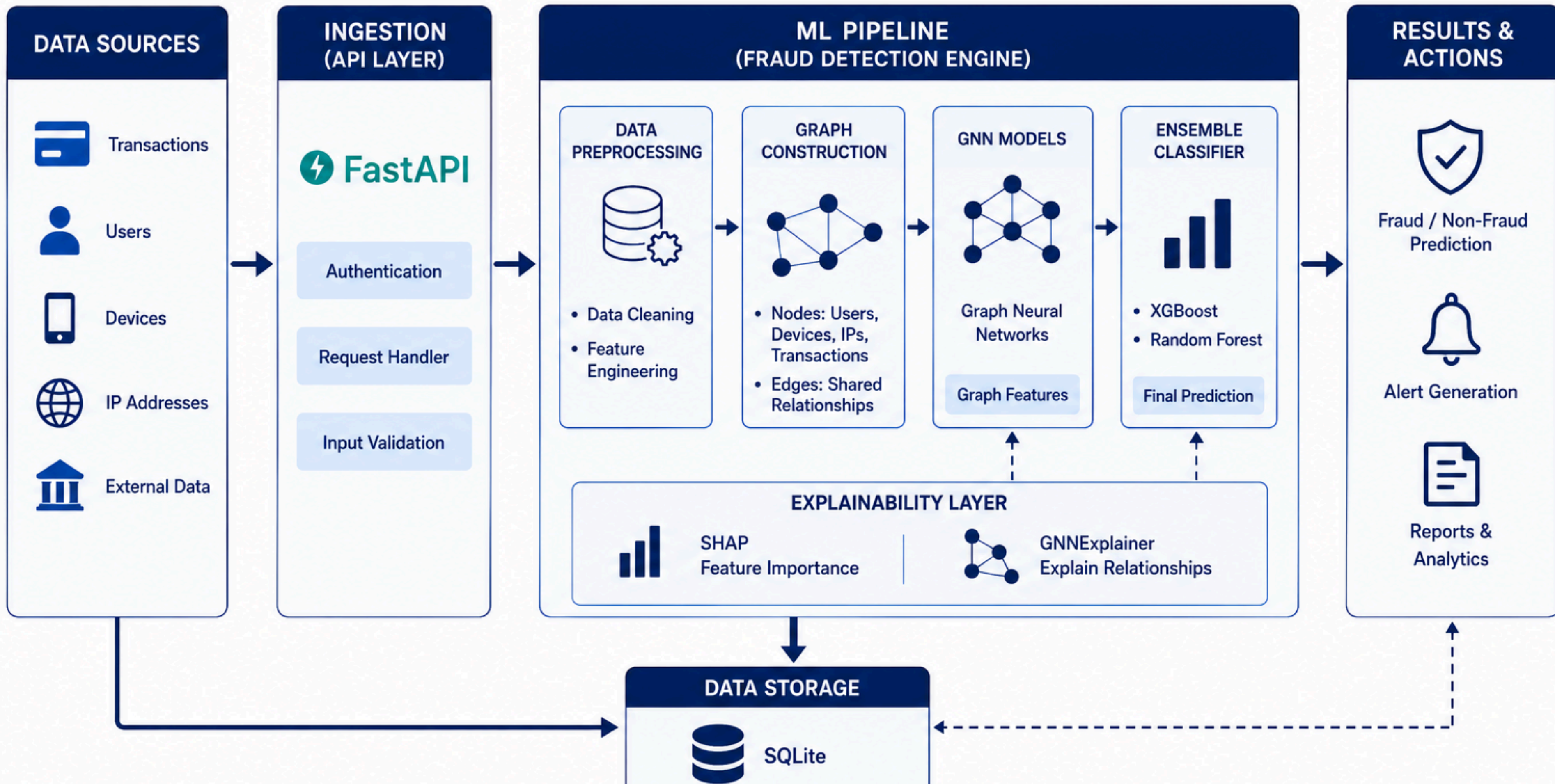


ALERTS &
NOTIFICATIONS



REPORTS

SYSTEM ARCHITECTURE



MODULES

CONCLUSION

CONCLUSION

REFERENCES